Multiparameter quantum metrology of incoherent point sources: Towards realistic superresolution

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We establish the multiparameter quantum Cramér-Rao bound for simultaneously estimating the centroid, the separation, and the relative intensities of two incoherent optical point sources using a linear imaging system. For equally bright sources, the Cramér-Rao bound is independent of their separation, which confirms that the Rayleigh resolution limit is just an artifact of the conventional direct imaging and can be overcome with an adequate strategy. For the general case of unequally bright sources, the amount of information one can gain about the separation falls to zero, but we show that there is always a quadratic improvement in an optimal detection in comparison with the intensity measurements. This advantage can be of utmost importance in realistic scenarios, such as observational astronomy.

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I. INTRODUCTION

The time-honored Rayleigh criterion [1] specifies the minimum separation between two incoherent optical sources using a linear imaging system. As a matter of fact, it is the size of the point spread function (PSF) [2] that determines the resolution: two points closer than the PSF width will be difficult to resolve due to the substantial overlap of their images.

Until recently, this Rayleigh criterion has been considered as a fundamental limit. Resolution can be improved only either by reducing the wavelength or by building higher numerical-aperture optics, thereby making the PSF narrower. Nonetheless, outstanding methods have been developed lately that can break the Rayleigh limit under special circumstances [3–12]. Though promising, these techniques are involved and require careful control of the source, which is not always possible, especially in astronomical applications.

Despite being very intuitive, the common derivation of the Rayleigh limit is heuristic, and it is deeply rooted in classical optical technology [13]. Recently, inspired by ideas of quantum information, Tsang and coworkers [14–17] have revisited this problem using the Fisher information and the associated Cramér-Rao lower bound (CRLB) to quantify how well the separation between two point sources can be estimated. When only the intensity at the image is measured (the basis of all the conventional techniques), the Fisher information falls to zero as the separation between the sources decreases and the CRLB diverges accordingly; this is known as the Rayleigh curse [14]. However, when the Fisher information of the complete field is calculated, it stays constant and so does the CRLB, revealing that the Rayleigh limit is not essential to the problem.

These remarkable predictions prompted a series of experimental implementations [18–21] and further generalizations [22–28], including the related question of source localization [29–31]. All this previous work has focused on the estimation of the separation, taking for granted a highly symmetric configuration with equally bright sources. Here we approach the issue in a more realistic scenario, where the sources may have unequal intensities. This involves the simultaneous estimation of separation, centroid, and intensities. Typically, when estimating multiple parameters, there is a trade-off in how well different parameters may be estimated: when the estimation protocol is optimized from the point of view of one parameter, the precision with which the remaining ones can be estimated deteriorates.

Actually, we show that including intensity in the estimation problem does lead to a reduction in the information for unbalanced sources. However, the information available in an optimal measurement still surpasses that of a conventional direct imaging scheme by a significant margin at small separations. This suggests possible applications, for example, in observational astronomy, where sources typically have small angular separations and can have large differences in brightness.

II. FORMULATION OF THE PROBLEM

Let us first set the stage for our simple model. We assume quasimonochromatic paraxial waves with one specified polarization and one spatial dimension, x denoting the image-plane coordinate. The corresponding object-plane coordinates can be obtained via the lateral magnification of the system, which we take to be linear spatially invariant. We emphasize that these assumptions are standard in imaging sciences [2] and they do not alter our main conclusions.

To facilitate possible generalizations, we phrase what follows in a quantum parlance. A wave of complex amplitude $U(x)$ can thus be assigned to a ket $|U\rangle$, such that $U(x) = \langle x|U\rangle$, where $|x\rangle$ is a vector describing a pointlike source at x.

The system is characterized by its PSF, which represents its normalized intensity response to a point source. We denote this PSF by $I(x) = |\langle x|\Psi\rangle|^2 = |\Psi(x)|^2$, so that $\Psi(x)$ can be interpreted as the amplitude PSF.

Two mutually incoherent point sources, of different intensities and separated by a distance $\mathfrak{s}$, are imaged by that system.

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The signal can be represented as a density operator

$$\rho_{\boldsymbol{\theta}} = \mathfrak{q}\, \rho_{+} + (1-\mathfrak{q})\, \rho_{-}, \tag{1}$$

where $\mathfrak{q}$ and $1-\mathfrak{q}$ are the intensities of the sources, with the proviso that the total intensity is normalized to unity. In addition, we have defined $\rho_{\pm} = |\Psi_{\pm}\rangle\langle\Psi_{\pm}|$ and the x-displaced PSF states are

$$\langle x|\Psi_{\pm}\rangle = \langle x - \mathfrak{s}_0 \mp \mathfrak{s}/2|\Psi\rangle = \Psi(x-\mathfrak{s}_0 \mp \mathfrak{s}/2), \tag{2}$$

so that they are symmetrically located around the geometric centroid $\mathfrak{s}_0 = \frac{1}{2}(x_{+} + x_{-})$. Note that

$$|\Psi_{\pm}\rangle = \exp[-i(\mathfrak{s}_0 \pm \mathfrak{s}/2)P]|\Psi\rangle, \tag{3}$$

where P is the momentum operator, which generates displacements in the x variable. As in quantum mechanics, it acts as a derivative $P = -i\partial_x$. These spatial modes are not orthogonal ($\langle\Psi_{-}|\Psi_{+}\rangle \neq 0$), so they cannot be separated by independent measurements.

We stress that, by its very nature, a pointlike source is specified by a Dirac delta complex amplitude, which results in a spatially coherent optical field. This justifies treating the spatial degrees of freedom of both signal components as pure states [32].

The density matrix $\rho_{\boldsymbol{\theta}}$ gives the normalized mean intensity

$$\rho_{\boldsymbol{\theta}}(x) = \mathfrak{q}\, |\Psi(x-\mathfrak{s}_0-\mathfrak{s}/2)|^2 + (1-\mathfrak{q})\, |\Psi(x-\mathfrak{s}_0+\mathfrak{s}/2)|^2 \tag{4}$$

and depends on the centroid $\mathfrak{s}_0$, the separation $\mathfrak{s}$, and the relative intensities of the sources $\mathfrak{q}$. This is indicated by the vector $\boldsymbol{\theta} = (\mathfrak{s}_0,\mathfrak{s},\mathfrak{q})^t$.

III. MULTIPARAMETER CRAMÉR-RAO LOWER BOUNDS

The task is to estimate the values of $\boldsymbol{\theta}$ through the measurement of some observables on $\rho_{\boldsymbol{\theta}}$. In turn, a quantum estimator $\widehat{\boldsymbol{\theta}}$ for $\boldsymbol{\theta}$ is a selfadjoint operator representing a proper measurement followed by data processing performed on the outcomes. Such a parameter estimation implies uncertainties for the measured values, which cannot be avoided.

In this multiparameter estimation scenario, the central quantity is the quantum Fisher information matrix (QFIM) [33]. This is a natural generalization of the classical Fisher information, which is a mathematical measure of the sensitivity of an observable quantity to changes in its underlying parameters. However, the QFIM is optimized over all the possible quantum measurements. It is defined as

$$Q_{\alpha\beta}(\boldsymbol{\theta}) = \tfrac{1}{2}\,\mathrm{Tr}(\rho_{\boldsymbol{\theta}}\{L_\alpha,L_\beta\}), \tag{5}$$

where the Greek indices run over the components of the vector $\boldsymbol{\theta}$ and $\{\cdot,\cdot\}$ denotes the anticommutator. Here L_α stands for the symmetric logarithmic derivative [34] with respect the parameter θ_α, represented implicitly as

$$\tfrac{1}{2}(L_\alpha\rho_{\boldsymbol{\theta}} + \rho_{\boldsymbol{\theta}}L_\alpha) = \partial_\alpha\rho_{\boldsymbol{\theta}}, \tag{6}$$

with $\partial_\alpha = \partial/\partial\theta_\alpha$.

Upon writing $\rho_{\boldsymbol{\theta}}$ in its eigenbasis, $\rho_{\boldsymbol{\theta}} = \sum_n \lambda_n|\lambda_n\rangle\langle\lambda_n|$, the QFIM can be concisely expressed as [35]

$$Q_{\alpha\beta}(\boldsymbol{\theta}) = 2\sum_{m,n}\frac{1}{\lambda_m+\lambda_n}\langle\lambda_m|\partial_\alpha\rho_{\boldsymbol{\theta}}|\lambda_n\rangle\langle\lambda_n|\partial_\beta\rho_{\boldsymbol{\theta}}|\lambda_m\rangle, \tag{7}$$

and the summation extends over m,n with $\lambda_m+\lambda_n \neq 0$.

The QFIM is a distinguishability metric on the space of quantum states and leads to the multiparameter quantum CRLB [36,37] for a single detection event:

$$\mathrm{Cov}(\widehat{\boldsymbol{\theta}}) \geqslant Q^{-1}(\boldsymbol{\theta}), \tag{8}$$

where $\mathrm{Cov}_{\alpha\beta}(\widehat{\boldsymbol{\theta}}) = \mathbb{E}[(\widehat{\theta_\alpha}-\theta_\alpha)(\widehat{\theta_\beta}-\theta_\beta)]$ refers to the covariance matrix for a locally unbiased estimator $\widehat{\boldsymbol{\theta}}$ of the quantity $\boldsymbol{\theta}$ and $\mathbb{E}[Y]$ is the expectation value of the random variable Y. The above inequality should be understood as a matrix inequality. In general, we can write $\mathrm{Tr}[G\,\mathrm{Cov}(\widehat{\boldsymbol{\theta}})] \geqslant \mathrm{Tr}[G\,Q^{-1}(\boldsymbol{\theta})]$, where G is some positive cost matrix, which allows us to asymmetrically prioritise the uncertainty cost of different parameters.

In using Eq. (8) we implicitly assume that the measurement is repeated N times, and this number is large. In astronomy, this corresponds to collecting many photons from the measured source, which is always the case, even for the stars at the detection limit of the telescope. For N detections, the right-hand side of Eq. (8) is divided by N. Because of this simple proportionality, factor N can be omitted and always reintroduced into the final expressions if needed.

The individual parameter θ_α can be estimated with a variance satisfying $\mathrm{Var}(\widehat{\theta_\alpha}) \geqslant (Q^{-1})_{\alpha\alpha}(\boldsymbol{\theta})$, and a positive operator-valued measurement (POVM) attaining this accuracy is given by the eigenvectors of L_α. Unlike for a single parameter, the collective bound is not always saturable: the intuitive reason for this is incompatibility of the optimal measurements for different parameters [38].

If the operators L_α corresponding to the different parameters commute, there is no additional difficulty in extracting maximal information from a state on all parameters simultaneously. If they do not commute, however, this does not immediately imply that it is impossible to simultaneously extract information on all parameters with precision matching that of the separate scenario for each. As discussed in a number of papers [39–41] the multiparameter quantum CRLB can be saturated provided

$$\mathrm{Tr}(\rho_{\boldsymbol{\theta}}[L_\alpha,L_\beta]) = 0, \tag{9}$$

with $[\cdot,\cdot]$ being the commutator. Then, optimal measurements can be found by optimizing over the classical Fisher information, as the QFIM is an upper bound for the former quantity. This can be efficiently accomplished by global optimization algorithms (see the Appendix). For our particular case, it is easy to see that the condition (9) is fulfilled whenever the PSF is real, $\Psi(x)^* = \Psi(x)$, which will be assumed henceforth. Additionally, for $N \gg 1$ the classical CRLB of the optimal measurement (i.e., the quantum CRLB) can be saturated with maximum-likelihood estimators. In the following we may thus replace all inequalities with their corresponding upper and lower bounds.

To proceed further, we note that the density matrix $\rho_{\boldsymbol{\theta}}$ is, by definition, of rank 2. The QFIM reduces then to the simpler form

$$\begin{aligned} Q_{\alpha\beta} = &-\frac{3}{\lambda_1}\langle\lambda_1|\partial_\alpha\rho_{\boldsymbol\theta}|\lambda_1\rangle\langle\lambda_1|\partial_\beta\rho_{\boldsymbol\theta}|\lambda_1\rangle \\ &-\frac{3}{\lambda_2}\langle\lambda_2|\partial_\alpha\rho_{\boldsymbol\theta}|\lambda_2\rangle\langle\lambda_2|\partial_\beta\rho_{\boldsymbol\theta}|\lambda_2\rangle \\ &+4\left(1-\frac{1}{\lambda_1}-\frac{1}{\lambda_2}\right)\langle\lambda_1|\partial_\alpha\rho_{\boldsymbol\theta}|\lambda_2\rangle\langle\lambda_2|\partial_\beta\rho_{\boldsymbol\theta}|\lambda_1\rangle \\ &+\frac{4}{\lambda_1}\langle\lambda_1|\partial_\alpha\rho_{\boldsymbol\theta}\partial_\beta\rho_{\boldsymbol\theta}|\lambda_1\rangle+\frac{4}{\lambda_2}\langle\lambda_2|\partial_\alpha\rho_{\boldsymbol\theta}\partial_\beta\rho_{\boldsymbol\theta}|\lambda_2\rangle. \end{aligned} \tag{10}$$

The derivatives involved in this equation can be easily evaluated; the result reads

$$\begin{aligned} \partial_{\mathfrak{s}_0}\rho_{\boldsymbol\theta} &= i[\rho_{\boldsymbol\theta},P], \\ \partial_{\mathfrak{s}}\rho_{\boldsymbol\theta} &= \tfrac{i}{2}(\mathfrak{q}[\rho_+,P]-(1-\mathfrak{q})[\rho_-,P]), \\ \partial_{\mathfrak{q}}\rho_{\boldsymbol\theta} &= \rho_+-\rho_-. \end{aligned} \tag{11}$$

To complete the calculation it proves convenient to write the two nontrivial eigenstates of $\rho_{\boldsymbol\theta}$ in terms of nonorthogonal component states $|\Psi_\pm\rangle$: $|\lambda_{1,2}\rangle = a_{1,2}|\Psi_+\rangle + b_{1,2}|\Psi_-\rangle$, where $a_{1,2}$ and $b_{1,2}$ are easy-to-find yet complicated functions of the separation and the intensities and whose explicit form is of no relevance for our purposes here. Substituting this and Eq. (11) into Eq. (10), and after a lengthy calculation, we obtain a compact expression for the QFIM:

$$Q = 4\begin{pmatrix} p^2+4\mathfrak{q}(1-\mathfrak{q})\wp^2 & (\mathfrak{q}-1/2)p^2 & -iw\wp \\ (\mathfrak{q}-1/2)p^2 & p^2/4 & 0 \\ -iw\wp & 0 & \dfrac{1-w^2}{4\mathfrak{q}(1-\mathfrak{q})} \end{pmatrix}. \tag{12}$$

This is our central result. The QFIM depends only on the following quantities:

$$\begin{aligned} w &\equiv \langle\Psi_\pm|\Psi_\mp\rangle = \langle\Psi|\exp(i\mathfrak{s}P)|\Psi\rangle, \\ p^2 &\equiv \langle\Psi_\pm|P^2|\Psi_\pm\rangle = \langle\Psi|P^2|\Psi\rangle, \\ \wp &\equiv \pm\langle\Psi_\pm|P|\Psi_\mp\rangle = \langle\Psi|\exp(i\mathfrak{s}P)P|\Psi\rangle. \end{aligned} \tag{13}$$

Notice carefully that p^2 is solely determined by the shape of the PSF, whereas both w and $\wp$ (which is purely imaginary) depend on the separation $\mathfrak{s}$.

IV. RESULTS

In what follows, rather than the variances themselves, we will use the inverses

$$H_\alpha = \frac{1}{\mathrm{Var}(\theta_\alpha)}, \tag{14}$$

usually called the precisions [42]. In this way, we avoid potential divergences as $\mathfrak{s}\to 0$.

The QFIM (12) nicely shows the interplay between various signal parameters. First, notice that Q is independent of the centroid, as might be expected. Second, for equally bright sources, $\mathfrak{q}=1/2$, the measurement of separation $\mathfrak{s}$ is uncorrelated with the measurements of the remaining parameters and we have $H_{\mathfrak{s}}(\mathfrak{q}=1/2)=p^2$, a well-known result, and the Rayleigh curse is lifted [18]. This happy situation does not hold for unequal intensities $\mathfrak{q}\neq 1/2$; now the separation is correlated with the centroid (via the intensity term $\mathfrak{q}-1/2$), and the centroid is correlated with the intensity (via p^2). This can be intuitively understood: unequal intensities result in asymmetrical images and finding the centroid is no longer a trivial task. This asymmetry, in turn, depends on the relative brightness of the two components. Hence, all the three parameters become correlated and, as we shall see, having separation-independent information about $\mathfrak{s}$ is no longer possible.

Indeed, by inverting the QFIM we immediately get

$$\begin{aligned} H_{\mathfrak{s}_0} &= 4\mathscr{Q}^2\frac{\wp^2+p^2(1-w^2)}{1-w^2}, \\ H_{\mathfrak{s}} &= p^2\mathscr{Q}^2\frac{\wp^2+p^2(1-w^2)}{\mathscr{Q}^2\wp^2+p^2(1-w^2)}, \\ H_{\mathfrak{q}} &= \frac{4}{\mathscr{Q}^2}\frac{\wp^2+p^2(1-w^2)}{\wp^2+p^2}, \end{aligned} \tag{15}$$

where $0<\mathscr{Q}^2\equiv 4\mathfrak{q}(1-\mathfrak{q})\leqslant 1$. Obviously, $H_{\mathfrak{s}}(\mathfrak{q})\leqslant H_{\mathfrak{s}}(\mathfrak{q}=1/2)=p^2$ and $\lim_{\mathfrak{q}\to 0,1} H_{\mathfrak{s}}(\mathfrak{q})=0$, which demonstrates that resolving two highly unequal sources is difficult, even at the quantum limit.

The instance of large brightness differences and small separations is probably the most interesting regime encountered, e.g., in exoplanet observations. We first expand the $\mathfrak{s}$-dependent quantities:

$$\begin{aligned} w(\mathfrak{s}) &= \langle\Psi|e^{i\mathfrak{s}P}|\Psi\rangle \simeq 1-\tfrac{1}{2}p^2\mathfrak{s}^2+\tfrac{1}{24}p^4\mathfrak{s}^4, \\ p(\mathfrak{s}) &= \langle\Psi|Pe^{i\mathfrak{s}P}|\Psi\rangle \simeq ip^2\mathfrak{s}-i\tfrac{1}{6}p^4\mathfrak{s}^3, \end{aligned} \tag{16}$$

where $p^4=\langle\Psi|P^4|\Psi\rangle$ is the fourth moment of the PSF momentum. Then, as $\mathfrak{s}\ll 1$, we get (for $0<\mathscr{Q}<1$)

$$\begin{aligned} H_{\mathfrak{s}_0} &\simeq \mathscr{Q}^2\,\mathrm{Var}(\hat{P}^2)\,\mathfrak{s}^2, \\ H_{\mathfrak{s}} &\simeq \frac{\mathscr{Q}^2}{4(1-\mathscr{Q}^2)}\,\mathrm{Var}(\hat{P}^2)\,\mathfrak{s}^2, \\ H_{\mathfrak{q}} &\simeq \frac{1}{\mathscr{Q}^2}\mathrm{Var}(\hat{P}^2)\,\mathfrak{s}^4. \end{aligned} \tag{17}$$

The PSF enters these expressions through the variance of P^2: $\mathrm{Var}(\hat{P}^2)=\mathbb{E}[(P^2-p^2)^2]$. This leaves room for optimization, provided the PSF can be controlled. For a fixed PSF, the information about all three parameters vanish with $\mathfrak{s}\to 0$, unless $\mathfrak{q}=1/2$. Since exactly balanced sources are uncommon, the information about very small separations always drops to near zero and the Rayleigh curse is unavoidable. However, significant improvements of the optimal measurement schemes over the standard intensity detection are still possible.

To illustrate this point let us consider a Gaussian response $\langle x|\Psi\rangle=(2\pi)^{1/4}\exp(-x^2/4)$ of unit width, which will serve from now on as our basic unit length. We compare the quantum limit given by (12) with that given by the classical Fisher information for the direct intensity measurement. We assume no prior knowledge about any of the three parameters.

Figure 1 plots information about separation $H_{\mathfrak{s}}$ for different relative intensities $\mathfrak{q}$. Unbalanced intensities make both optimal and intensity detection go to zero for small separations;

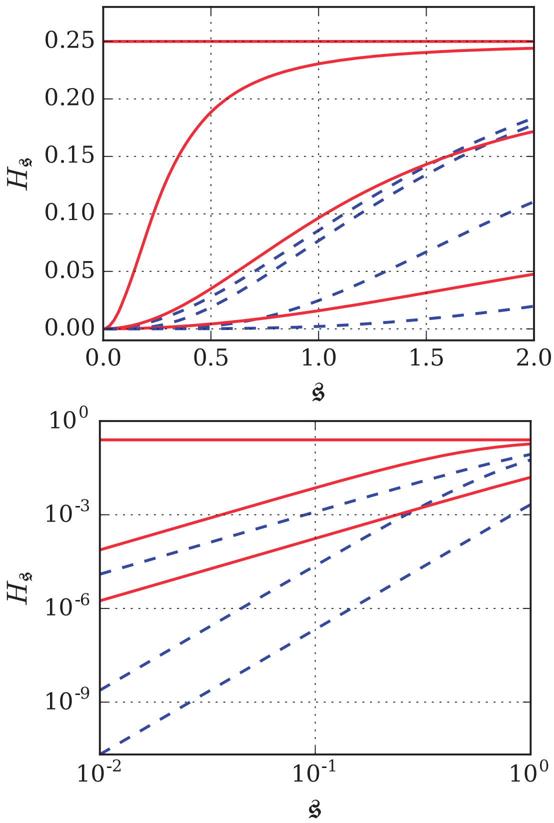

FIG. 1. Upper panel: Precision $H_{\mathfrak{s}}$ in the separation $\mathfrak{s}$ as inferred by optimal (red solid lines) and direct (blue broken lines) detections, for different relative intensities of the two sources. The values of $\mathfrak{q}$, from top to bottom, are 0.5, 0.45, 0.3, and 0.1. Notice that the performance of the optimal detection is rather sensitive to small deviations from equal brightness over a wide range of separations. Lower panel: Precisions visualized on a logarithmic scale. Slopes of straight lines translate to the powers of $\mathfrak{s}$. The values of $\mathfrak{q}$ are, from top to bottom, 0.5, 0.4, and 0.1.

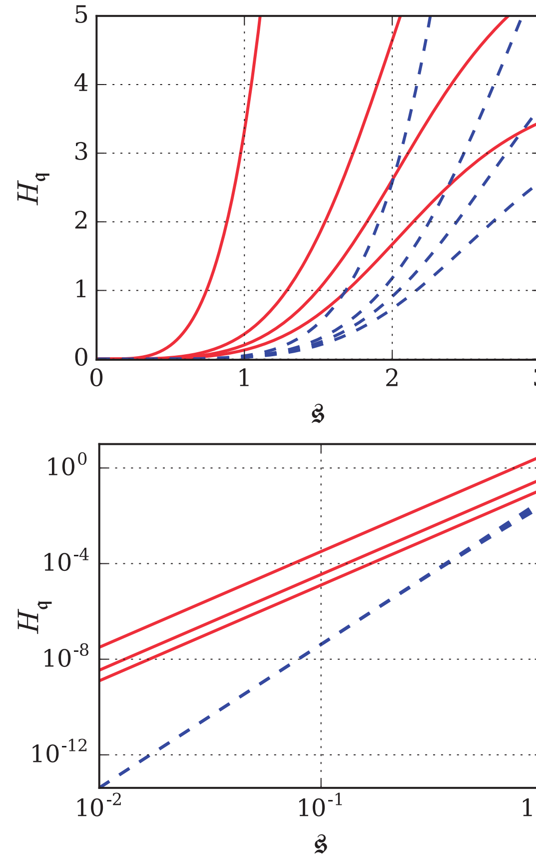

FIG. 2. Upper panel: Precision $H_{\mathfrak{q}}$ in the relative intensity $\mathfrak{q}$ as inferred by optimal (red solid lines) direct (blue broken lines) detections, for different relative intensities of the two sources. The values of $\mathfrak{q}$, from bottom to top are 0.5, 0.2, 0.1, and 0.01. Lower panel: Precisions visualized on a logarithmic scale. Slopes of straight lines translate to the powers of $\mathfrak{s}$. The three blue broken lines cannot be distinguished on this scale. The values of $\mathfrak{q}$ are, from bottom to top, 0.5, 0.2, and 0.01.

however, the former at a much slower rate. Hence, the ratio of optimal to intensity information increases with decreasing separations, regardless of whether the sources are balanced.

The reason becomes obvious with the same data visualized on the logarithmic scales, as shown in the lower panel of Fig. 1. In the region of $\mathfrak{s} \ll \sigma$, we can discern two regimes of importance. For balanced sources, $H_{\mathfrak{s}}^{\rm opt} \propto 1$ and $H_{\mathfrak{s}}^{\rm int} \propto \mathfrak{s}^2$. For unbalanced sources, $H_{\mathfrak{s}}^{\rm opt} \propto \mathfrak{s}^2$, as we have seen, and $H_{\mathfrak{s}}^{\rm int} \propto \mathfrak{s}^4$. In consequence, there is always a factor of $\mathfrak{s}^{-2}$ improvement of the optimal detection over the standard one, irrespective of the true values of the signal parameters. In practice, this means that when we already are well below the Rayleigh limit, if we decrease the separation 10 times, about 10 000 times more photons must be detected with a CCD camera to keep the accuracy of the measurement, while only 100 times more would suffice for optimal measurement. This amounts to saving 99% of detection time with the optimal detection scheme.

Figure 2 presents a similar comparison, but now concerning the precision $H_{\mathfrak{q}}$ in the relative intensity. Here optimal information and intensity information always scale as $\mathfrak{s}^4$ and $\mathfrak{s}^6$, respectively, and the same $\mathfrak{s}^{-2}$ gain in performance appears. Notice the reversed ordering of curves with $\mathfrak{q}$, meaning that now the information increases with increasing intensity difference, which reveals the complementarity between these magnitudes. Also notice that the broken lines converge as we approach $\mathfrak{s} = 0$. It can be shown that the leading term for intensity detection for small separations is $\mathfrak{q}$-independent, in contrast to the optimal detection, which displays a strong $H_{\mathfrak{q}}^{\rm opt} \propto \mathfrak{q}^{-1}$ dependence for $\mathfrak{q} \ll 1/2$. This highlights the advantage of an optimal detection scheme for astronomical observations. For example, more than a quarter of catalogued binary systems consist of stars that differ in brightness by more than an order of magnitude [43], and the darkest known exoplanet is three orders of magnitude dimmer than its host star in the infrared [44].

Finally, Fig. 3 shows that for highly unbalanced sources, the advantage of the quantum detection over the direct intensity imaging extends to larger separations. For balanced sources ($\mathfrak{q} = 0.5$) direct imaging drops below 50% of the corresponding quantum limit at $\mathfrak{s} \simeq 1.3\sigma$. This border value grows to $\mathfrak{s} \simeq 2.3\sigma$ at $\mathfrak{q} = 0.1$ and further to $\mathfrak{s} \simeq 3.8\sigma$ at $\mathfrak{q} = 0.01$. This means one can benefit from quantum detection well above the

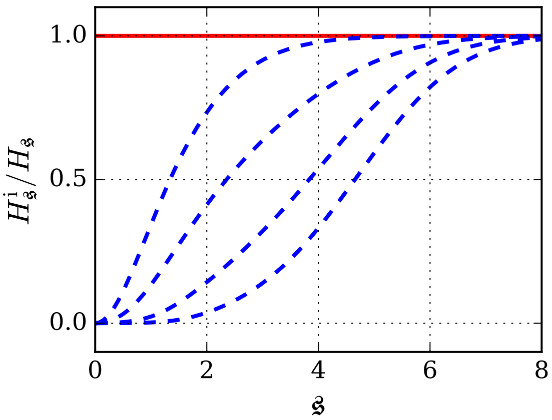

FIG. 3. Precision about separation provided by the direct intensity imaging normalized by the corresponding quantum limit. Notice how the gap between the optimal detection (red solid line) and direct imaging (blue broken lines) widens with the two sources getting more unequal. The values of $\mathfrak{q}$ are, from bottom to top, 0.001, 0.01, 0.1, and 0.5.

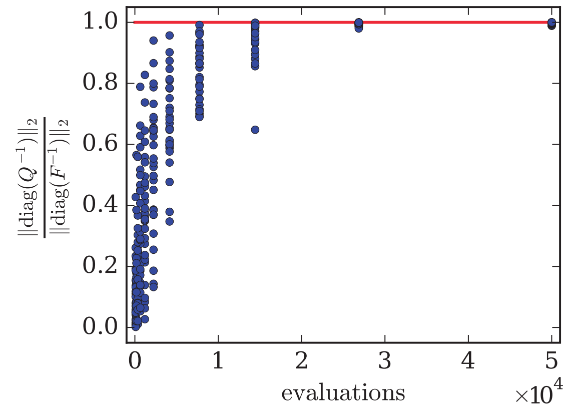

FIG. 4. POVM optimization. The POVM performance (blue dots) approaches the quantum limit (red line) in the course of optimization. POVM is optimized for 30 different randomly generated signal states. For each signal state we vary the total number of objective function evaluations between 100 and 50 000 times.

Rayleigh limit provided the two signal components are highly unequal, which is often the case.

V. CONCLUSIONS

In summary, we have presented a comprehensive analysis of the ultimate precision bounds for estimating the centroid, the separation, and the relative intensities of two pointlike incoherent sources. For equally bright sources, the quantum Fisher information remains constant, which translates into the fact that the Rayleigh limit is not essential and can be lifted. On the other hand, for unequally bright sources, the information about very small separations always drops to near zero and the Rayleigh curse is unavoidable. Nonetheless, significant improvements can still be expected with optimal detection schemes. Work along these lines is in progress.

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APPENDIX: OPTIMAL DETECTION

In principle, analytical techniques to obtain a measurement saturating the quantum CRLB are at hand, but they apply only to pure states [45]. For mixed states, we can find such an optimal detection by numerical maximization of the classical Fisher information matrix. It is useful to work in a suitable computational basis formed by a set of orthonormal modes denoted by $|n\rangle$, with $n = 1,2,\ldots,d$. The sampled basis wave functions $V_{kn} = \langle x_k|n\rangle$ readily transform the signal state (1) and signal state derivatives (11) between the x representation (2) and our computational n representation. Since we deal with real-valued amplitude PSFs only, the basis wave functions can be also taken as real.

We parametrize the measurement by a real vector, which is required by most black-box optimization routines. We construct the measurement POVM of size M as $\Pi_j = |u_j\rangle\langle u_j|$ using $M-1$ rank-one projectors on a set of orthonormal states $|u_j\rangle$ $(j = 1,\ldots,M-1)$. This together with the complement $\Pi_M = \mathbb{1} - \sum_k^{M-1}\Pi_k$ makes a valid POVM of size M in a Hilbert space of dimension d.

In the computational basis, such a POVM is fully specified by a set of $M-1$ orthonormal d-dimensional vectors

$$\langle n|u_1\rangle,\ldots,\langle n|u_{M-1}\rangle. \tag{A1}$$

Taking a $d \times M-1$ real matrix Z, such a set is provided by the basis of the range space and can be easily generated using the SVD decomposition $Z = USV^\dagger$, with

$$\langle n|u_1\rangle = U_{n1},\ldots,\langle n|u_{M-1}\rangle = U_{nM-1}. \tag{A2}$$

In this way, any real-valued $d \times M-1$ matrix generates a POVM. This matrix, in turn, is a reshaped $d(M-1)$ dimensional real vector z establishing a convenient vector parametrization of the measurement $\Pi_j(z)$.

Quantum Fisher information is approached by minimizing a suitable function of classical Fisher information matrix

$$F_{\alpha\beta}(z) = \sum_{j=1}^{M}\frac{\partial_\alpha p_j(z)\partial_\beta p_j(z)}{p_j(z)}, \tag{A3}$$

where $p_j(z) = \mathrm{Tr}[\rho_{\boldsymbol{\theta}}\Pi_j(z)]$ are the probabilities of the M measurement outputs. One option is to minimize the L_2 norm of the diagonal elements of the Fisher matrix inverse

$$\min_z \|\,\mathrm{diag}[F(z)^{-1}]\|_2. \tag{A4}$$

Global optimization tools are needed to solve such a nonconvex problem [46]. Notice that estimating three independent parameters requires a POVM with at least $M = 4$ outputs and $3d$ parameters to optimize. This is within the grasp of currently available numerical tools and hardware.

Figure 4 shows typical results obtained for a Gaussian PSF in the Hermite-Gauss computational basis $d = 6$ subject to measurements with $M = 4$ outputs. The signal parameters are randomly sampled from the parameter space $\mathfrak{s}_0 \in [-1,1]$, $\mathfrak{s} \in [0.05,0.5]$, and $\mathfrak{q} \in [0.1,0.5]$ exploring the region of interest well below the Rayleigh limit. The problem is solved by the global optimization package BlackBoxOptim for Julia [47] using the default differential evolution optimizer. In most cases a few tens of thousand evaluations of Fisher information matrix are enough to secure two significant digits of the quantum limit. This translated to less than 10˜s computation time on a standard i7-7700 CPU. We stress that the resulting optimal measurements, although challenging, can be feasibly implemented in the laboratory.

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